

Review

Navigating AKT-ivity across cellular compartments

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Phosphoinositide (PIP)-mediated AKT signaling is essential for cellular homeostasis because it orchestrates crucial processes such as metabolism, survival, proliferation, and motility. Dysregulation of this pathway drives various pathologies, particularly cancer. Although cytosolic activation of AKT has been extensively studied, its emerging roles in the nucleus have gained attention over the past decade. The complexities of AKT compartmentalization and associated functional mechanisms remain largely unexplored. At the plasma membrane, AKT activation occurs at specialized microdomains and cell–cell junctions where it influences polarity, adhesion, and migration. In endosomes, PIPs coordinate intracellular trafficking and cytoskeletal organization with AKT signaling. Protein scaffolds refine AKT signal specificity by assembling complexes. In the nucleus, AKT interacts with the p53–PIP signalosome and specific kinases to regulate oncogenesis and chemoresistance. This review explores PIP-driven spatial regulation of AKT across cellular compartments, emphasizing its role in cellular responses and oncogenesis. Understanding AKT compartmentalization mechanisms provides valuable insights into cancer biology and novel therapeutic strategies.

Overview of AKT signaling

The AKT pathway, a cornerstone of cellular signaling, has evolved significantly since the identification of the *v-Akt* oncogene from the AKT8 retrovirus in 1987 [1]. By 1991 the cellular homolog of *v-AKT* was independently cloned by three research groups who named it c-AKT [2], related to A- and C-kinase (RAC) [3], and protein kinase B (PKB) [4], respectively. AKT is a serine/threonine kinase and a central member of the AGC kinase family, named from its founding members, the protein kinases A, G, and C. Mammalian genomes encode three isoforms – AKT1, AKT2, and AKT3 – that have distinct tissue distributions. AKT1 and AKT2 are ubiquitous, whereas AKT3 is abundant in brain, kidney, and heart [5,6]. Despite high homology in their catalytic domains, differences in the N-terminal pleckstrin homology (PH) and C-terminal regulatory domains contribute to isoform-specific functions [5]. Dysregulated AKT signaling is implicated in diseases, especially cancer, where it drives oncogenesis and progression [7].

The canonical model situates AKT at the cell membrane where its activation requires **phosphatidylinositol (PI or PtdIns)** (see [Glossary](#)) 3-kinase (PI3K) signaling ([Figure 1](#)). PI3K generates the **phosphoinositide (PIP) phosphatidylinositol 3,4,5-trisphosphate** [PtdIns(3,4,5)P₃] that recruits AKT via its pleckstrin homology (PH) domain for phosphorylation by phosphoinositide-dependent protein kinase 1 (PDK1) at Thr308 and by mechanistic target of rapamycin kinase complex 2 (mTORC2) at Ser473, leading to activation of the kinase and downstream signaling [8]. In its inactive state, AKT adopts an autoinhibitory 'PH-in' conformation. Binding of PtdIns(3,4,5)P₃ to the PH domain, via key basic residues such as Lys14, Arg23, and Arg25, triggers a conformational shift to the 'PH-out' state, relieving autoinhibition and enabling phosphorylation by upstream kinases such as PDK1 [9,10]. This review explores AKT signaling

Highlights

The spatial regulation of AKT across cellular compartments, including the plasma membrane, endosomes, protein scaffolds, and the nucleus, orchestrates cellular processes.

Specialized microdomains at the plasma membrane enable precise AKT activation and control cell polarity, adhesion, and migration.

Endosomes function as signaling platforms for sustained AKT activity, and RAB5 and phosphoinositides (PIPs) play crucial roles.

Protein scaffolds such as IQGAP1, AKI1, and CLIPR-59/CLIP3 enhance AKT signaling specificity and efficiency across compartments.

Nuclear AKT, modulated by the p53–PIP signalosome or specific kinases, drives oncogenesis and chemoresistance.

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dynamics across cellular compartments, offering insights into its roles in cancer and therapeutic opportunities.

Recent research reveals that AKT can be activated beyond the plasma membrane (PM) [10], including at endosomes [11,12], **protein scaffolds** [13–15], and in the nucleus [16,17]. This spatial regulation underpins the ability of AKT to orchestrate cellular processes, including proliferation, apoptosis resistance, and metabolism [8,18].

AKT signaling at the plasma membrane

The PM regulates cellular processes such as proliferation, survival, and metabolism, and serves as the canonical site for the PI3K/AKT signaling pathway (Figure 2). Lipid rafts are cholesterol- and sphingolipid-enriched PM microdomains that participate in numerous cellular functions [19]. Fluorescence correlation spectroscopy studies demonstrate that lipid rafts are essential for recruiting AKT to the PM following PtdIns(3,4,5)P₃ production, and disruption of rafts inhibits this recruitment [20]. Furthermore, fluorescence resonance energy transfer (FRET)-based AKT activity reporter (AKTAR) studies confirm that AKT is preferentially activated within lipid rafts [21]. Treatment with the cholesterol-depleting agent methyl- β -cyclodextrin ablates phosphorylation of lipid raft-associated AKT1, and myristoylated AKT1 localizes to cholesterol-rich membranes, suggesting a potential cholesterol–AKT1 interaction [22]. Similarly, PDK1 activation occurs within lipid rafts upon growth factor stimulation, whereas phosphatase and tensin homolog (PTEN), a negative PI3K/AKT pathway regulator, localizes predominantly to non-raft regions [23]. Recent evidence further suggests that AKT undergoes palmitoylation that enables its direct insertion into the lipid bilayer and enhances its proximity to PtdIns(3,4,5)P₃-bound PDK1, thereby promoting its activation [24]. Disruption of this compartmentalization impairs AKT activity, deactivates mTORC1/p70 ribosomal protein S6 kinase (p70S6K) signaling, and activates the forkhead box protein O3 (FOXO3) transcription factor, ultimately leading to apoptosis [25]. These findings underscore the crucial role of lipid raft-mediated compartmentalization in modulating PI3K/AKT signaling.

Focal adhesion (FA) sites are dynamic multi-protein complexes that are essential for anchoring cells to the extracellular matrix (ECM) and enabling bidirectional signal transmission [26]. Integrin activation, driven by binding to extracellular ligands or intracellular proteins such as talin, paxillin, and vinculin, is essential for recruiting FA components and the maturation of these structures [26].

PIP signaling, particularly by PtdIns(4,5)P₂ generated by PIPK γ , is pivotal in regulating FA dynamics. PtdIns(4,5)P₂ activates talin and vinculin that link integrins to actin filaments, thereby modulating cytoskeletal organization and cell motility [8,18].

FA proteins also regulate PM protrusions such as lamellipodia, filopodia, and invadopodia which drive cell motility [27]. These structures reflect actin cytoskeletal remodeling at the leading edge of the cell orchestrated mainly by PIPs and RHO family small GTPases, including RAS-related C3 botulinum toxin substrate 1 (RAC1) and cell division control protein 42 homolog (CDC42) [28,29]. At the leading edge of migrating cells, PtdIns(4,5)P₂ is converted by PI3K into PtdIns(3,4,5)P₃, which can be further processed into PtdIns(3,4)P₂ by SRC homology 2 (SH2) domain-containing inositol 5-phosphatase 1 (SHIP1) [30]. Alternatively, PtdIns(3,4)P₂ can be produced by PI3KC2 β from PtdIns4P and plays a role in FA disassembly during changes in the cellular environment [31]. Both PtdIns(3,4)P₂ and PtdIns(3,4,5)P₃ bind to the PH domain of AKT for its activation [32,33], implicating AKT in FA turnover. Indeed, AKT modulates FA disassembly by phosphorylating focal adhesion kinase (FAK) at serine residues [34,35]. The localization of PTEN, a phosphatase targeting both PtdIns(3,4,5)P₃ and PtdIns(3,4)P₂, to the rear of migrating cells likely offsets AKT activation [30]. The dynamic assembly and disassembly of FAs, tightly linked to AKT activity, are essential for cell migration, invasion, and survival.

Glossary

Central tenet of lipid biology: a key concept in lipid signaling is that lipid effectors are closely coupled to lipid-metabolizing enzymes. This spatial coupling ensures that newly generated lipid signals remain localized, thereby allowing rapid and efficient signal transduction.

Phosphatidylinositol (PI or PtdIns): a membrane phospholipid precursor for generating key signaling molecules. It comprises a glycerol backbone, two fatty acid chains, and an inositol ring with six hydroxyl groups. The inositol head group can undergo phosphorylation at the 3rd, 4th, and 5th positions, and this forms the basis for PIP signaling.

Phosphoinositides (PIPs): phosphorylated derivatives of PI/PtdIns that are generated through dynamic enzymatic modifications of the inositol ring. This family includes seven distinct isomers: phosphatidylinositol 3-phosphate (PtdIns3P), PtdIns4P, PtdIns5P, PtdIns(3,4)P₂, PtdIns(3,5)P₂, PtdIns(4,5)P₂, and PtdIns(3,4,5)P₃. Each isomer has specific subcellular distributions and functions that regulate diverse processes such as intracellular signaling, membrane identity, and trafficking. PIP metabolism is governed by tightly regulated networks of lipid kinases, phosphatases, and phospholipases that ensure the spatial and temporal precision of cellular signaling.

PIP signaling: PIPs mediate intracellular signaling by serving as docking platforms for effector proteins with specialized lipid-binding domains, such as pleckstrin homology (PH) domains, PHOX homology (PX) domains, and FYVE (FAB1, YOTB, VAC1, and EEA1) domains. A prominent example is the phosphatidylinositol 3-kinase (PI3K)/AKT pathway, where PtdIns(3,4,5)P₃ recruits and activates AKT to govern cell proliferation, survival, and metabolism. Other PIP species also contribute to specific functions: PtdIns4P supports Golgi trafficking whereas PtdIns(4,5)P₂ regulates actin dynamics, plasma membrane (PM) signaling, and nuclear events. In addition, PtdIns(4,5)P₂ acts as a precursor for second messengers such as diacylglycerol (DAG) and inositol trisphosphate (IP₃) to amplify their signaling potential.

Lamellipodia formation has been shown to occur via the AKT pathway because the PI3K/AKT inhibitor LY294002 effectively blocks its induction [36]. RAC1 and CDC42 contain AKT substrate motifs, and RAC1 is phosphorylated by AKT at Ser71 *in vitro* [37]. AKT pathway inhibition reduces RAC1 phosphorylation, whereas AKT2 activation enhances RAC1 activity [38]. Phosphorylated AKT colocalizes with RAC1 and CDC42 at the leading edge of the cell where active RAC1 and CDC42 stimulate AKT activity, forming a positive feedback loop that promotes cell motility [39,40].

Once activated, RAC1 and CDC42 drive actin remodeling through the actin-related protein 2/3 (ARP2/3) complex via the suppressor of the cAMP receptor (SCAR)/Wiskott–Aldrich syndrome protein (WASP) family verprolin homolog (WAVE) protein [41–43]. Within the WAVE regulatory complex (WRC), RAC1 binding induces conformational changes that activate the ARP2/3 complex and mediate actin nucleation [44]. This cascade establishes reinforcing cycles that can locally restrict protrusions through proteins such as CYRI-B and ARPIN that interact with the WRC and the ARP2/3 complexes, respectively [45,46]. The ARP2/3 complex may further promote cancer progression by modulating PI3K/AKT signaling [47]. As lamellipodia develop, integrins bind to ECM ligands and trigger signaling cascades via FAK and paxillin, which recruit guanine nucleotide exchange factors (GEFs) such as GIT1 and β -PIX for RAC1 [48]. This activation promotes actin polymerization through the ARP2/3 complex and further elongates FAs.

The activation of the SCAR/WAVE complex and the recruitment of the ARP2/3 complex rely on AKT [49]. AKT and the ARP2/3 complex appear to function codependently because AKT colocalizes with ARP2/3 in lamellipodia and membrane ruffles through ubiquitin-like protein 4A (UBL4A)-mediated binding [50]. Inhibition of either ARP2/3 or the actin network disrupts both AKT membrane translocation and activation, suggesting that actin filament assembly is essential for recruiting AKT, and AKT in turn supports the activity of the ARP2/3 complex at the leading edge of cell protrusions.

AKT also functions at cell–cell junctions, where N-cadherin mediates adherens junction formation between adjacent cells and activates the PI3K/AKT pathway [51]. Exposure to tumor-treating fields (TTFields) enhances homophilic N-cadherin interactions, thereby increasing intercellular adhesion, amplifying PI3K/AKT signaling, and promoting the inactivation of BCL2-associated agonist of cell death (BAD), ultimately supporting cell survival. Notably, pharmacological inhibition of PI3K sensitizes cancer cells to TTFields, suggesting a potential combinatorial strategy for future cancer therapies [51].

Moreover, AKT is active at invadopodia, where caveolin-1 acts as a mechanosensor activated by low shear stress [52,53]. Upon activation, caveolin-1 translocates and clusters at the PM, initiating the PI3K/AKT–mTOR pathway [53]. This cascade upregulates membrane-type 1 matrix metalloproteinase (MT1-MMP) expression and enhances ECM degradation, facilitating invadopodia formation and promoting metastasis [53].

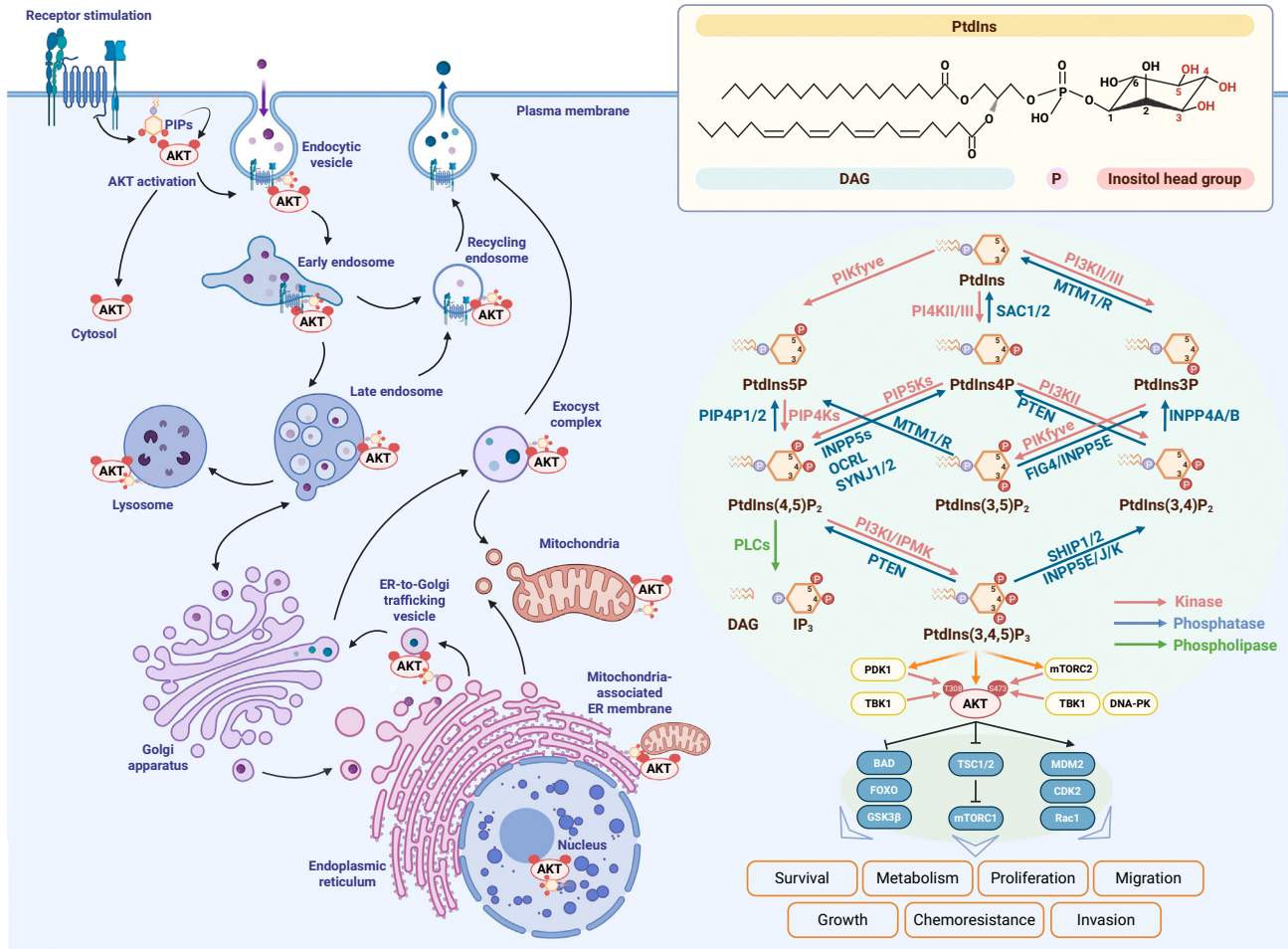
AKT signaling at endosomes

After activation at the PM, AKT signaling extends to the endosomal compartment (Figure 3). In the endocytic network, AKT plays a key role early in stimulation. RAB5, a GTPase that regulates endocytosis and endosome fusion, depends on PI3K for activation, and subsequently affects AKT kinase activity [54]. Inhibition of PI3K by wortmannin disrupts endocytosis and endosome fusion, highlighting AKT as a primary effector downstream of PI3K in endocytosis [54]. Further studies confirmed that RAS-driven endocytosis requires both AKT and RAB5, solidifying their roles in mediating the effects of RAS on endocytosis [55]. In addition, RAB5 interacts with insulin receptor substrate 1 (IRS1) and coordinates insulin-like growth factor (IGF)–AKT–mTOR signaling in muscle regeneration [56].

PI transfer proteins (PITPs):

specialized lipid-binding proteins that mediate the transfer of phosphatidylinositol (PI) and other phospholipids between membrane compartments or from membranes to specific protein targets. PITPs are essential for maintaining phosphoinositide (PIP) homeostasis and ensuring the precise spatial and temporal regulation of PIP signaling. PITPs are classified into two main groups: class I PITPs (e.g., PITP α and PITP β) and class II PITPs (e.g., phosphatidylinositol transfer protein cytoplasmic 1, PITPNC1), phosphatidylinositol transfer protein membrane-associated 1 (PITPNM1), and PITPNM2. These lipid transfer proteins function in a countercurrent mode of action where class I PITPs primarily shuttle PI and phosphatidylcholine (PC) whereas class II PITPs transfer PI and phosphatidic acid (PA).

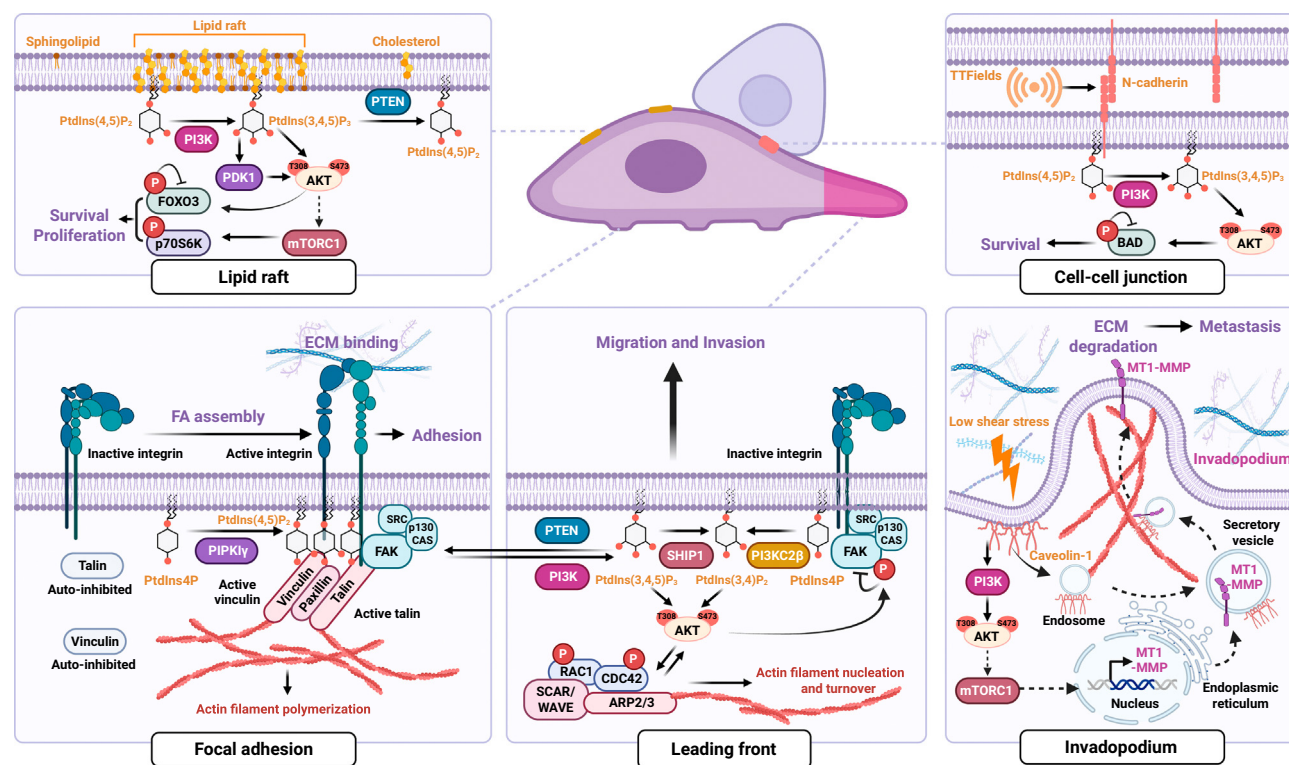
Protein scaffolds: these organize signaling pathways by bringing together enzymes, substrates, and regulators into functional complexes. This spatial proximity enhances signal fidelity, specificity, and speed, and thus enables cells to respond efficiently to stimuli.



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Figure 1. AKT signaling across cellular compartments and underlying phosphoinositide (PIP) metabolism. The left panel illustrates the spatial distribution of AKT signaling, often in association with PIPs, across key cellular compartments, including the plasma membrane (PM), endosomes/lysosomes, nucleus, cytosol, mitochondria, mitochondria-associated endoplasmic reticulum (ER) membranes (MAMs), ER-to-Golgi trafficking vesicles, and the exocyst complex. The right panel depicts the metabolism of phosphatidylinositol (PI or PtdIns) and its phosphorylated derivatives – collectively termed PIPs – which regulate AKT activation. The structure of PI/PtdIns is shown as a diacylglycerol (DAG) linked to an inositol ring via a phosphate group. The inositol ring of PI contains hydroxyl groups at positions 3, 4, and 5 which can be phosphorylated to generate seven major PIP species: PtdIns3P, PtdIns4P, PtdIns5P, PtdIns(3,4)P₂, PtdIns(3,5)P₂, PtdIns(4,5)P₂, and PtdIns(3,4,5)P₃. These species are dynamically interconverted by specific lipid-modifying enzymes, including kinases (phosphatidylinositol 3-kinases, PI3Ks; phosphatidylinositol 4-kinases, PI4Ks; phosphoinositide kinase for five position containing a FYVE finger, PIKFYVE; phosphatidylinositol-5-phosphate 4-kinases, PIP5Ks; phosphatidylinositol-4-phosphate 5-kinases, PIP5Ks; and inositol polyphosphate multikinase, IPMK), phosphatases (suppressors of actin 1/2, SAC1/2; factor-induced gene 4, FIG4; myotubularins, MTM1/MTMRs; phosphatidylinositol-4,5-bisphosphate 4-phosphatases 1/2, PIP4P1/2; inositol polyphosphate-5-phosphatases, INPP5s; oculocerebrorenal syndrome of Lowe protein, OCRL; synaptotagmins 1/2, SYNJ1/2; inositol polyphosphate-4-phosphatases A/B, INPP4A/B; SH2 domain-containing inositol 5-phosphatases 1/2, SHIP1/2; and phosphatase and tensin homolog, PTEN), and phospholipases (phospholipase C family, PLCs). In the canonical PI3K/AKT pathway, PtdIns(3,4,5)P₃ generated at the cell membrane recruits AKT via its pleckstrin homology (PH) domain. Subsequent phosphorylation of AKT at Thr308 by phosphoinositide-dependent protein kinase-1 (PDK1), and at Ser473 by mechanistic target of rapamycin complex 2 (mTORC2), fully activates the kinase. These residues can also be phosphorylated independently of PtdIns(3,4,5)P₃ by kinases such as TANK-binding kinase 1 (TBK1) and DNA-dependent protein kinase (DNA-PK). Activated AKT regulates a broad network of downstream effectors including BCL2-associated agonist of cell death (BAD), tuberous sclerosis complex 1/2 (TSC1/2), mouse double minute 2 homolog (MDM2), forkhead box protein O (FOXO), glycogen synthase kinase 3β (GSK3β), mechanistic target of rapamycin complex 1 (mTORC1), cyclin-dependent kinase 2 (CDK2), and RAS-related C3 botulinum toxin substrate 1 (RAC1). Through these interactions, AKT orchestrates crucial cellular processes such as survival, metabolism, proliferation, growth, migration, invasion, and chemoresistance. This figure was created with BioRender.

There are two distinct populations of RAB5-positive early endosomes: one containing the canonical RAB5 effector early endosome antigen 1 (EEA1) and the other harboring adaptor protein phosphotyrosine interacting with PH domain and leucine zipper 1 (APPL1) [57]. These



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Figure 2. AKT signaling at the plasma membrane. This figure illustrates how AKT signaling orchestrates cellular processes such as migration and invasion through its action at lipid rafts, cell-cell junctions, focal adhesions (FAs), the leading front of migrating cells, and invadopodia. At lipid rafts – cholesterol- and sphingolipid-enriched microdomains – the phosphatidylinositol 3-kinase (PI3K)/AKT signaling pathway is activated. PI3K catalyzes the conversion of phosphatidylinositol 4,5-bisphosphate [PtdIns(4,5)P₂] to PtdIns(3,4,5)P₃, whereas phosphatase and tensin homolog (PTEN) dephosphorylates PtdIns(3,4,5)P₃ back to PtdIns(4,5)P₂. Phosphoinositide-dependent protein kinase 1 (PDK1) is activated within lipid rafts upon growth factor stimulation and phosphorylates AKT at Thr308, and mechanistic target of rapamycin complex 2 (mTORC2) phosphorylates AKT at Ser473, leading to full activation. Activated AKT promotes cell survival and proliferation via downstream effectors such as mechanistic target of rapamycin complex 1 (mTORC1) and p70 ribosomal protein S6 kinase (p70S6K), as well as through inhibition of forkhead box protein O3 (FOXO3). At cell-cell junctions, exposure to tumor-treating fields (TTFs) enhances N-cadherin-mediated homophilic interactions, increasing intercellular adhesion and amplifying PI3K/AKT signaling. This promotes the phosphorylation and inactivation of BCL2-associated agonist of cell death (BAD), thereby supporting cell survival. At FAs, integrin activation is driven by extracellular matrix (ECM) binding or by intracellular proteins such as talin, paxillin, and vinculin. FA components including SRC, p130CAS, and focal adhesion kinase (FAK) are recruited, and phosphatidylinositol-4-phosphate 5-kinase type I γ (PIPKI γ) generates PtdIns(4,5)P₂, which activates talin and vinculin to link integrins to actin filaments, thereby modulating cytoskeletal organization and motility. PtdIns(4,5)P₂ is further converted into PtdIns(3,4,5)P₃ by PI3K, and then into PtdIns(3,4)P₂ by SH2 domain-containing inositol 5-phosphatase 1 (SHIP1). PtdIns(3,4)P₂ is also synthesized by the class II PI3K isoform PI3KC2 β from PtdIns4P and facilitates FA disassembly. Both PtdIns(3,4)P₂ and PtdIns(3,4,5)P₃ bind to the pleckstrin homology (PH) domain of AKT and enable its activation. Activated AKT regulates FA dynamics by phosphorylating FAK and other substrates. PTEN is localized to the rear of migrating cells where it counteracts AKT activation to maintain front-rear polarity. At the leading front of migrating cells, integrin engagement triggers signaling cascades via FAK and paxillin, which activate RAS-related C3 botulinum toxin substrate 1 (RAC1) and cell division control protein 42 homolog (CDC42). These GTPases stimulate actin cytoskeletal remodeling through the suppressor of cAMP receptor/WASP family verprolin homologous protein (SCAR/WAVE) complex and the actin-related protein 2/3 (ARP2/3) complex. AKT colocalizes with and phosphorylates RAC1 and CDC42 at the leading edge, creating a positive feedback loop that promotes motility. At invadopodia, under low shear stress, caveolin-1 functions as a mechanosensor that clusters at the plasma membrane and activates the PI3K/AKT–mTORC1 axis. This signaling cascade induces membrane-type 1 matrix metalloproteinase (MT1-MMP) expression, enhances ECM degradation, and promotes invadopodia formation and metastatic progression. This figure was created with BioRender.

endosomes exhibit distinct distributions that are influenced by the actin cytoskeleton and the nutritional status of the cell [58]. Both AKT and phospho-AKT interact with EEA1 [59]. EEA1-positive endosomes act as crucial sites for AKT activation in response to agonists such as angiotensin II, epidermal growth factor (EGF), and insulin [11,59]. AKT translocates to and is activated within these endosomes, and EEA1 expression is required for AKT phosphorylation at Thr308 and Ser473, as well as for the phosphorylation of downstream targets mTOR and S6 kinase, although

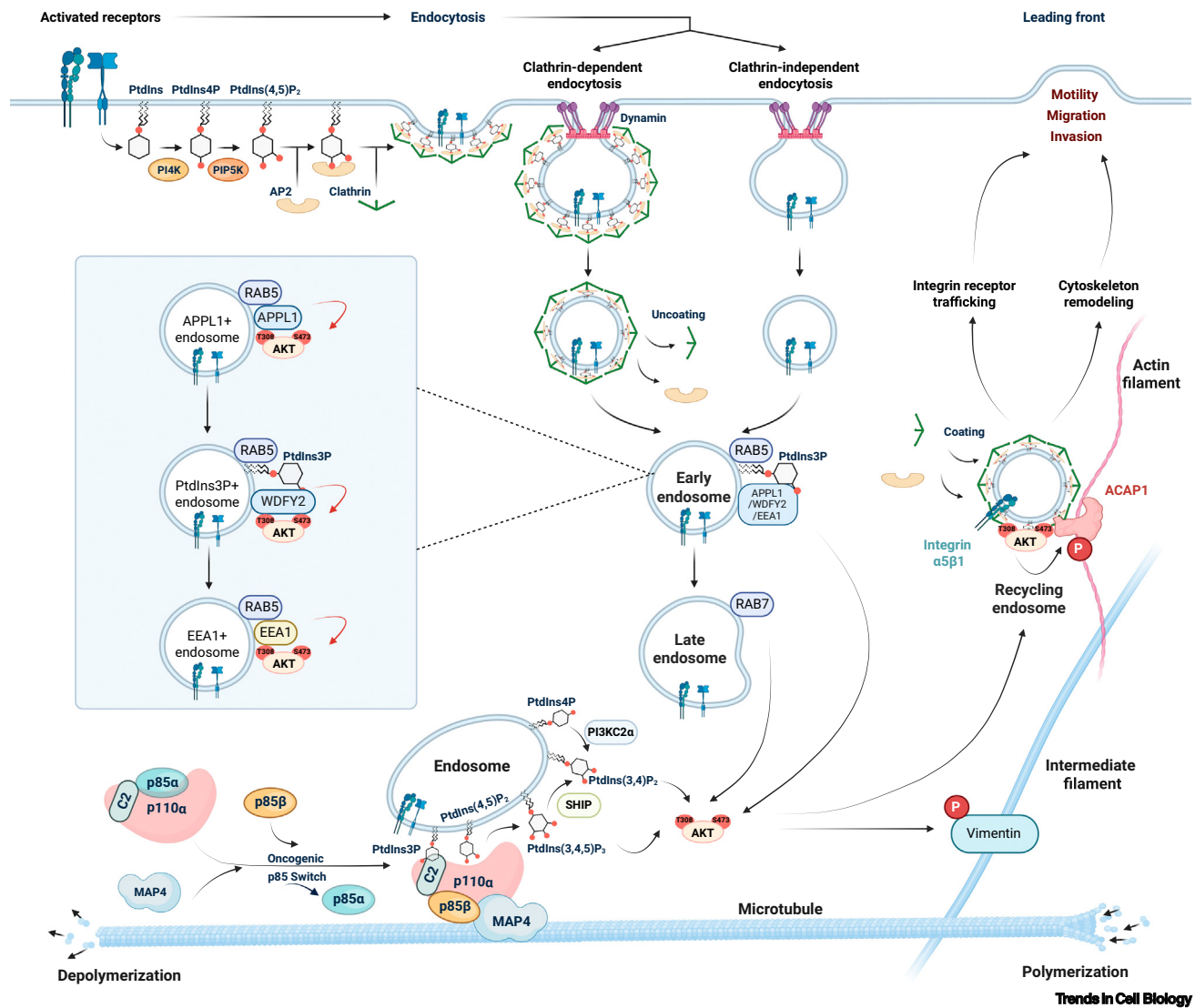


Figure 3. AKT signaling at endosomes. This figure illustrates the multifaceted roles of AKT signaling at endosomes in regulating endocytosis, membrane trafficking, recycling, and cellular dynamics including migration and invasion. Upon receptor activation, endocytosis is initiated through either clathrin-dependent or clathrin-independent pathways. In clathrin-dependent endocytosis, phosphatidylinositol 4,5-bisphosphate [PtdIns(4,5)P₂] is synthesized at membrane invaginations via the sequential activation of phosphatidylinositol 4-kinases (PI4Ks) and phosphatidylinositol-4-phosphate 5-kinases (PIP5Ks) at the plasma membrane. This lipid recruits adaptor protein complex 2 (AP-2), facilitating the assembly of clathrin-coated pits. Membrane scission is then mediated by the GTPase dynamin, leading to the formation of endocytic vesicles where AKT signaling is engaged. RAS-related protein RAB5, a small GTPase that is crucial for early endosome fusion, is activated via phosphatidylinositol 3-kinase (PI3K) and supports AKT kinase activity. RAB5-positive early endosomes are classified into three sequential populations based on their associated adaptor proteins: adaptor protein phosphotyrosine interacting with PH domain and leucine zipper 1 (APPL1), WD repeat and FYVE domain-containing protein 2 (WDFY2), and early endosome antigen 1 (EEA1). Each of these endosomal subtypes can recruit and activate AKT signaling context-dependently. Class I PI3K associates with microtubules via microtubule-associated protein 4 (MAP4), facilitating local production of PtdIns(3,4,5)P₃ and endosomal AKT activation. During oncogenic transformation, the regulatory subunit of the PI3K catalytic subunit p110α shifts from PI3K regulatory subunit p85α to p85β, and this enhances its recruitment to endosomes. The endosomal membrane is enriched in phosphoinositides – especially PtdIns(4,5)P₂ which is crucial for adaptor complex recruitment and serves as a substrate for PI3K-mediated generation of PtdIns(3,4,5)P₃, thereby activating AKT. In addition to its kinase function, AKT acts as a coadaptor with ARF-GAP with coiled-coil, ankyrin repeat, and PH domain-containing protein 1 (ACAP1) to regulate the recycling of cargo proteins such as α5β1 integrin to the leading edge of migrating cells. As endosomes mature into late endosomes and fuse with lysosomes, RAS-related protein RAB7 functions as a master regulator linking late endosomal trafficking to AKT signaling. RAB7 promotes AKT-mediated phosphorylation of vimentin, an intermediate filament protein that is involved in cytoskeletal dynamics. Through its coordination of integrin receptor recycling and cytoskeletal remodeling, AKT signaling along the endosomal track drives cellular motility, migration, and invasion. This figure was created with BioRender.

not for ERK1/2 activation [59], highlighting the pivotal role of EEA1-positive endosomes in mediating AKT signaling.

In neurons, AKT plays a crucial role in long-distance axonal endosomal transport that is vital for neuronal function [60]. AKT interacts with RAB5-positive endosomes that are crucial for transporting signaling molecules such as neurotrophins. Notably, APPL1, an adaptor protein for AKT, is localized in dendritic spines and axons of hippocampal neurons [60,61]. Overexpression of APPL1 enhances phospho-AKT levels at synaptic sites where it promotes spine and synapse formation [61]. FRET analysis shows that the interaction of AKT with APPL1 is endocytosis-dependent, suggesting that APPL1 links receptors to AKT and facilitates transport within axonal compartments [60]. Most APPL endosomes are precursors to PtdIns3P-positive endosomes, and PtdIns3P plays a crucial role in facilitating this conversion [62]. Depletion of PtdIns3P causes RAB5-positive endosomes to revert to the APPL1 stage, enhancing growth factor signaling and likely recruiting AKT via APPL1 [62].

Through PtdIns3P binding, WD repeat and FYVE domain-containing protein 2 (WDFY2) localizes to early endosomes upon insulin stimulation, and interacts with the insulin receptor (INSR) via its WD1-4 domain. WDFY2 recruits the INSR to endosomes and facilitates the recruitment of insulin receptor substrates 1 and 2 (IRS1/2) [63]. WDFY2 binds to AKT through its WD-repeat domains and enhances insulin-induced activation of AKT2 and promotes the phosphorylation of its substrates, including GSK-3 β and FOXO1 [63]. WDFY2 specifically colocalizes with endogenous AKT2, the primary isoform in insulin-stimulated glucose uptake, but not with AKT1 [64]. Activated WDFY2 forms a tetrameric complex that enhances FOXO1 phosphorylation at Ser253 and suppresses its anti-adipogenic activity. Depletion of WDFY2 impairs AKT2 phosphorylation, insulin-stimulated glucose transport, and adipogenic gene expression [64]. In *Wdfy2* knockout mice, insulin-induced AKT2/FOXO1 and AKT2/GSK-3 β signaling is disrupted in the liver, confirming the role of WDFY2-enriched endosomes in insulin signaling specificity [63].

Along microtubules, class I PI3K associates with microtubule-associated protein 4 (MAP4) and facilitates agonist-stimulated PtdIns(3,4,5)P₃ production and AKT activation at endosomes [11]. Upon agonist stimulation, PtdIns3P-enriched endosomes recruit the catalytic subunit of class I PI3K, p110 α , via the C2 domain, initiating PI3K/AKT signaling [12]. This process is amplified during oncogenesis through a switch in the regulatory subunit of p110 α from p85 α to p85 β that enhances its recruitment to endosomes via receptor tyrosine kinases [12], and this is a growing research area, especially in cancers with p85 α loss [65].

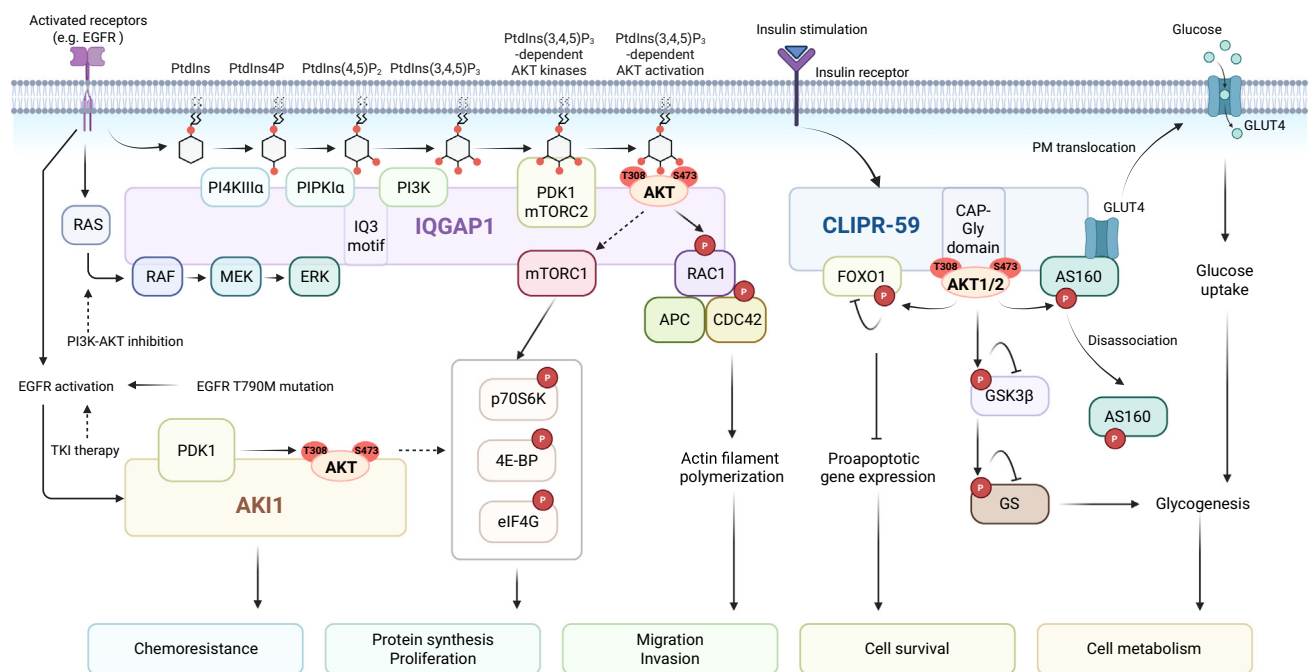
In addition to PtdIns3P, the endosomal membrane is enriched in PtdIns(4,5)P₂ that is essential for recruiting adaptor complexes such as adaptor protein 2 (AP-2), epsin, and monomeric adaptor protein with 180 kDa (AP180) during the formation of clathrin-coated endosomes [66]. Once clathrin-coated vesicles form, dynamin constricts the neck to separate the vesicle from the PM, a process that depends on PtdIns(4,5)P₂ to recruit dynamin [67]. Given the connection between PI3K/AKT signaling and endosomes, the enrichment of PtdIns(4,5)P₂ at clathrin-coated endosomes may serve as a substrate for PI3K-mediated conversion into PtdIns(3,4,5)P₃ and activation of AKT. AKT functions as a coadaptor in the clathrin-coated complex where it collaborates with ACAP1 (ARF-GAP with coiled-coil, ankyrin repeat and PH domain-containing protein 1), which controls actin remodeling and membrane trafficking, to facilitate the recycling of cargo proteins such as $\alpha 5 \beta 1$ integrin back to the PM [68], suggesting a role for AKT in endosome recycling.

Early endosomes mature into late endosomes and fuse with lysosomes for degradation in a tightly regulated process. RAB7, a small GTPase, sustains AKT-mediated survival signals during cell detachment; its silencing decreases phosphorylated AKT levels and thus reduces anchorage-

independent growth and promotes anoikis [69]. In addition, RAB7 regulates the phosphorylation of vimentin, an intermediate filament protein and mesenchymal marker in cancer cells [70]. Vimentin is phosphorylated at Ser39 by AKT, which enhances cellular motility and invasion while protecting against caspase-induced proteolysis [71]. RAB7 modulation of vimentin phosphorylation via AKT connects late endosomal trafficking with AKT signaling [70].

AKT signaling at protein scaffolds

Protein scaffolds play a crucial role in facilitating the assembly of the $\text{PtdIns}(3,4,5)\text{P}_3$ generation machinery and downstream AKT signaling components, exemplifying the **central tenet of lipid biology** by tethering lipid-metabolizing enzymes and lipid effectors, thereby ensuring efficient and accurate signal transduction (Figure 4). A prime example is the scaffold protein IQ



Trends in Cell Biology

Figure 4. AKT signaling at protein scaffolds. This figure illustrates the role of protein scaffolds in orchestrating AKT signaling by spatially integrating lipid-metabolizing enzymes, phosphoinositides (PIPs), and downstream effectors to enable efficient and compartmentalized signal transduction. Following activation of cell-surface receptors such as epidermal growth factor receptor (EGFR), the scaffold protein IQ motif-containing GTPase-activating protein 1 (IQGAP1) recruits a cascade of PIP-modifying enzymes – including phosphatidylinositol 4-kinase III α (PI4KIII α), phosphatidylinositol-4-phosphate 5-kinase type I α (PIP5K α), and phosphatidylinositol 3-kinase (PI3K) – to catalyze the stepwise generation of PIPs culminating in phosphatidylinositol 3,4,5-trisphosphate [$\text{PtdIns}(3,4,5)\text{P}_3$]. IQGAP1 also recruits AKT kinases such as phosphoinositide-dependent protein kinase 1 (PDK1) and mechanistic target of rapamycin complex 2 (mTORC2), leading to full AKT activation. Activated AKT subsequently phosphorylates downstream effectors such as mechanistic target of rapamycin complex 1 (mTORC1), which in turn activates p70 ribosomal protein S6 kinase (p70S6K), eukaryotic translation initiation factor 4E-binding protein (4E-BP), and eukaryotic translation initiation factor 4 γ (eIF4G), thus promoting protein synthesis and cell proliferation. In parallel, AKT signaling scaffolded on IQGAP1 also activates RAS-related C3 botulinum toxin substrate 1 (RAC1), cell division control protein 42 homolog (CDC42), and the adenomatous polyposis coli (APC) complex, likely through direct phosphorylation of RAC1 and CDC42 by AKT. This promotes actin filament polymerization, thereby enhancing cell migration and invasion. In addition, IQGAP1 scaffolds the RAS/RAF/mitogen-activated protein kinase kinase (MEK)/extracellular signal-regulated kinase (ERK) signaling pathway, particularly under PI3K/AKT inhibition, highlighting its dual role in coordinating parallel signaling networks. AKT kinase-interacting protein 1 (AKI1) is another protein scaffold that promotes AKT activation specifically downstream of EGFR stimulation. Its expression is amplified in tumors harboring the *EGFR* T790M mutation, where it facilitates complex formation between PDK1 and AKT, thereby sustaining AKT phosphorylation and supporting cancer cell proliferation and resistance to EGFR-tyrosine kinase inhibitor (TKI) therapy. CLIP-170-related 59 kDa protein (CLIPR-59) functions as a membrane scaffold that anchors AKT1/2 signaling machinery to the plasma membrane (PM) and the *trans*-Golgi network. It promotes the phosphorylation of key AKT substrates including AKT substrate of 160 kDa (AS160) and forkhead box protein O1 (FOXO1). Phosphorylation of AS160 causes its dissociation from glucose transporter type 4 (GLUT4), facilitating GLUT4 translocation and enhancing insulin-stimulated glucose uptake and glycogenesis through inhibition of glycogen synthase kinase 3 β (GSK3 β). Concurrently, phosphorylation of FOXO1 promotes its degradation, leading to repression of FOXO1-mediated proapoptotic gene expression. Through these mechanisms, CLIPR-59-anchored AKT signaling regulates metabolic homeostasis and cell survival. This figure was created with BioRender.

motif-containing GTPase-activating protein 1 (IQGAP1) which integrates PIPs and associated kinases to modulate the PI3K/AKT pathway [14]. IQGAP1 expression is elevated in several cancer types, including hepatocellular carcinoma, where 58% of tissue samples show increased IQGAP1 expression compared to adjacent normal tissue [72]. Elevated levels of IQGAP1 enhance the scaffolding of mTOR and AKT, resulting in increased AKT phosphorylation and activity [72]. In a mouse model of head and neck cancer (HNC) driven by the synthetic oral carcinogen 4-nitroquinoline 1-oxide (4NQO), IQGAP1 promotes carcinogenesis through PI3K/AKT scaffolding [73]. The level of PI3K signaling in 4NQO-induced HNC in *Iqgap1*^{-/-} mice was significantly reduced, supporting the hypothesis that IQGAP1 contributes to HNC at least partly through PI3K signaling. High IQGAP1 expression correlates with decreased survival, whereas elevated phosphorylation of ribosomal protein S6 (pS6), catalyzed by p70S6K upon mTORC1 activation, is associated with high IQGAP1 levels in HNC patients [73,74].

Beyond its role in the PI3K/AKT axis, IQGAP1 also scaffolds the RAS/RAF/MEK/ERK pathway that is activated in response to PI3K/AKT inhibition [13,75]. The IQ3 motif specifically mediates PI3K/AKT activation and enables the IQGAP1–PI3K complex to regulate cell proliferation, migration, and invasion independently of the MEK/ERK pathway [15]. IQGAP1 further interacts with mTORC1, leading to activation of the PI3K/AKT–mTOR cascade and sequential downstream signaling, including p70S6K, eukaryotic initiation factor 4G (eIF4G), and eukaryotic translation initiation factor 4E-binding protein (4E-BP), which govern protein synthesis and cell growth, and thus influences cell dynamics and motility [72]. In response to migratory cues, IQGAP1 also recruits a complex comprising adenomatous polyposis coli (APC) and two AKT substrates, RAC1 and CDC42 [37], linking actin filaments and microtubule dynamics during cell polarization and directional migration [76].

In addition to IQGAP1, another crucial protein scaffold in the PI3K/AKT signaling pathway is AKT kinase-interacting protein 1 (AKI1) which promotes PDK1–AKT complex formation and regulates AKT activation in a concentration-dependent manner [77]. In response to epidermal growth factor (EGF) stimulation, AKI1 recruits the EGF receptor (EGFR) to facilitate Thr308 phosphorylation of AKT, enhancing its activation. AKI1 does not influence insulin-induced AKT activation although it acts as a receptor-specific scaffold. In cancers such as pancreatic and lung, AKI1 is closely associated with elevated phospho-AKT levels and tumor cell survival [78,79]. Silencing AKI1 reduces phospho-AKT levels, induces apoptosis, and inhibits tumor growth, particularly in EGFR T790M gatekeeper mutant lung cancer cells. Furthermore, AKI1 is frequently expressed in *EGFR* mutant lung tumors, including tumors resistant to EGFR-tyrosine kinase inhibitor (TKI) therapy, and its knockdown enhances the effectiveness of next-generation EGFR-TKIs, making AKI1 a promising therapeutic target in cancer treatment [79].

CLIP-170-related 59 kDa protein (CLIPR-59, also known as CLIP3) is another protein scaffold that regulates AKT signaling compartmentalization within cells. It features three distinct structural domains: a Glu/Pro-rich domain, three ankyrin repeats, and two glycine-rich cytoskeleton-associated protein (CAP-Gly) domains [80]. Although CAP-Gly domains typically bind to microtubules and regulate their dynamics, CLIPR-59 does not exhibit microtubule-binding activity [81]. Instead, it associates with cellular membranes, such as the PM and *trans*-Golgi network, via a palmitoylated cysteine residue and serves as a scaffold for AKT signaling [82]. The CAP-Gly domains of CLIPR-59 interact with the kinase domain of AKT, whereas its ankyrin repeats scaffold key AKT substrates such as FOXO1 and AKT substrate of 160 kDa (AS160) [83]. AS160, a membrane-associated RAB GTPase-activating protein (GAP), is essential for GLUT4 membrane translocation, a crucial process for insulin-regulated glucose homeostasis [84]. Upon insulin stimulation, CLIPR-59 relocates to the cell membrane where it assembles AKT pathway components to phosphorylate AS160 at Ser318, Ser570, Ser588, Thr642, and Thr571, thereby suppressing its

GAP activity [85]. This leads to dissociation of AS160 from the CLIPR-59–GLUT4–AS160 complex and thus promotes GLUT4 membrane translocation and glucose transport [84]. CLIPR-59 can assemble AKT1 and AKT2, with a higher affinity for AKT2, aligning with the primary role of AKT2 in regulating insulin-stimulated glucose transport [83,86]. Active AKT phosphorylates and inhibits glycogen synthase kinase 3 β (GSK3 β), which regulates glucose homeostasis by phosphorylating and inactivating glycogen synthase (GS), thereby promoting increased glycogenesis [87]. Thus, CLIPR-59 plays a crucial role in fine-tuning AKT signaling and glucose metabolism.

Additional protein scaffolds also mediate AKT signaling, including connector enhancer of KSR 1 (CNK1), β -arrestin 2, casein kinase 2-interacting protein 1 (CKIP-1), c-JUN N-terminal kinase (JNK)-interacting protein 1 (JIP1), disabled homolog 2-interacting protein (DAB2IP), ezrin-radixin-moesin-binding phosphoprotein 50 (EBP50), heat-shock protein 90 (HSP90), plenty of SH3s (POSH), prohibitin (PHB), and receptor for activated C-kinase 1 (RACK1) [88]. These protein scaffolds are essential for the precise localization of signaling molecules, underscoring their importance in cellular signaling dynamics and the regulation of AKT activity.

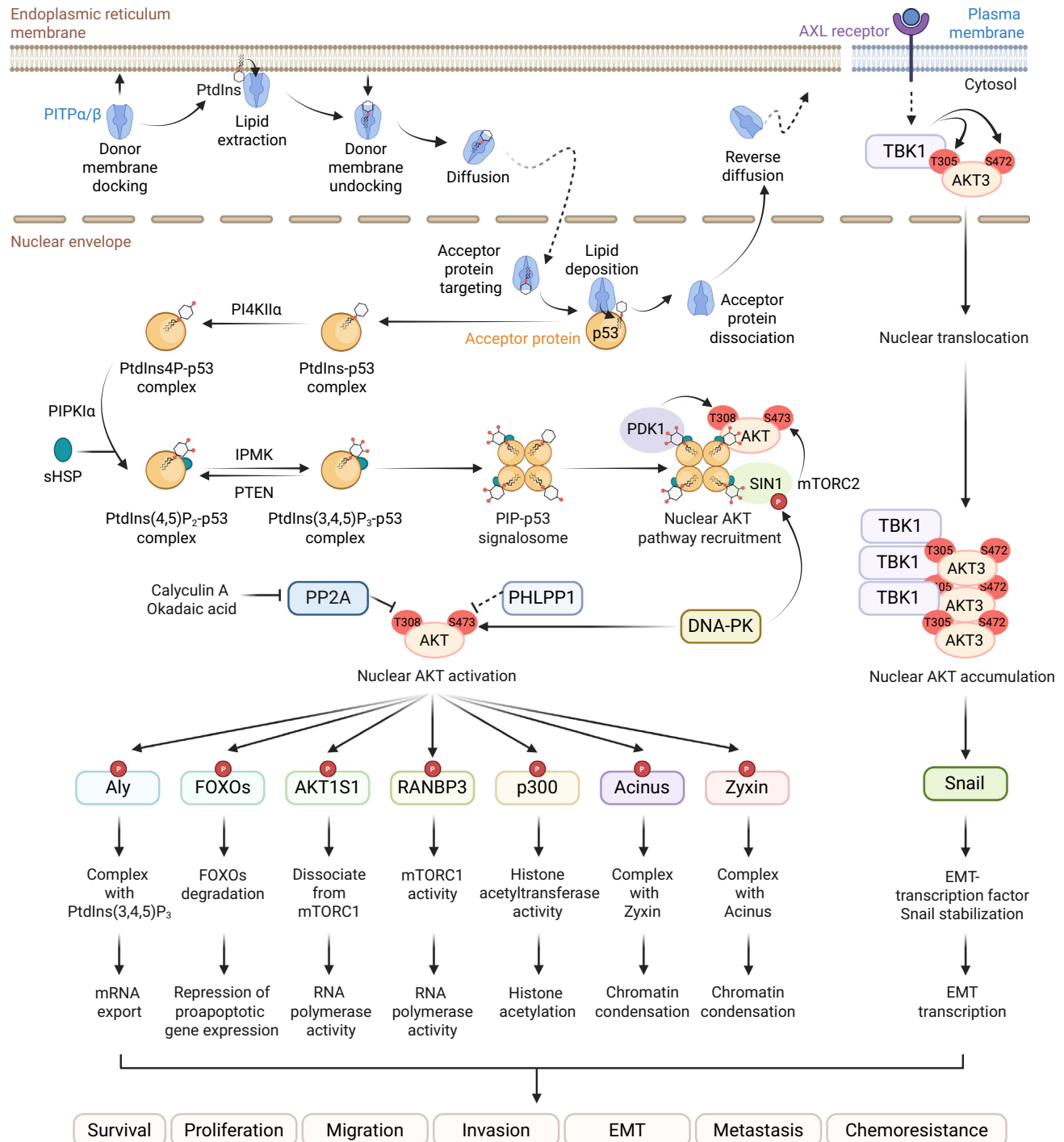
AKT signaling in the nucleus

AKT has emerged as a crucial player in nuclear signaling since the late 1990s. Studies at that time revealed that a variety of stimuli, including insulin, insulin-like growth factor 1 (IGF-1), hypoglycemia, fetal calf serum (FCS), nerve growth factor (NGF), and the F(ab')₂ fragment of anti-mouse IgG targeting the B cell receptor, activate AKT, leading to the nuclear translocation of AKT [89,90]. The absence of a nuclear localization sequence (NLS) in AKT suggests an alternative mechanism for translocation which recent research has begun to elucidate.

Protein kinases, phosphatases, and PIP signaling within the nucleus tightly control nuclear AKT signaling (Figure 5). PDK1, a master kinase that is crucial for activating AKT by phosphorylating it at Thr308, is present in the nucleus and shows enhanced nuclear localization in response to stimuli such as insulin, IGF-1, NGF, PDGF, and genotoxic stress [16,91–93]. Although lacking an NLS, PDK1 contains a nuclear export signal (NES) [93]. PDK1 forms a complex with the NLS-containing tyrosine phosphatase SHP-1 that facilitates its nucleocytoplasmic shuttling [94]. Phosphorylation at Ser396 near the NES of PDK1 is necessary for nuclear shuttling of PDK1 [92]. Immunofluorescence staining against phospho-Ser396 of PDK1 shows a specific correlation with the nuclear PtdIns(4,5)P₂ islet but not with the cytosolic PtdIns(4,5)P₂ signal under genotoxic stress [16], suggesting conversion of PtdIns(4,5)P₂ to PtdIns(3,4,5)P₃ near PDK1 may support nuclear AKT activation.

mTORC2 is the kinase complex that mediates the phosphorylation of AKT at Ser473, facilitating its full activation. It comprises multiple subunits: mTOR, DEPTOR, mLST8, which are shared with mTORC1, and three unique subunits specific to mTORC2 – RICTOR, mSIN1, and PROTOR [95], that are partially present in the nucleus and respond to cellular stimulation [16,96]. The PH domain of mSIN1 binds to PtdIns(3,4,5)P₃, promoting mTORC2 activation and influencing cellular localization [97]. PDK1, mTORC2, and AKT all require binding to PtdIns(3,4,5)P₃ for activation [10,98,99].

Although PI synthesis is restricted to the endoplasmic reticulum (ER) because key enzymes such as CDP-diacylglycerol synthase 1/2 (CDS1/2) and PI synthase (PIS) are ER membrane-integral proteins [100], **PI transfer proteins (PITPs)** and PIP-metabolizing enzymes enable the generation of nuclear PtdIns(3,4,5)P₃ [90,101,102]. Nuclear-localized PI3K that is responsible for PtdIns(3,4,5)P₃ production includes class I PI3K with catalytic subunits p110 α and p110 β and regulatory subunits p85 α and p85 β , as well as the nuclear-localized PI3K inositol polyphosphate multikinase (IPMK) [16,102–109]. In addition, the nuclear localization of the PtdIns(3,4,5)P₃



Trends in Cell Biology

Figure 5. AKT signaling in the nucleus. This figure illustrates the mechanisms by which AKT signaling is activated and functions within the nucleus. Following phosphatidylinositol (PI or PtdIns) synthesis at the endoplasmic reticulum (ER), phosphatidylinositol transfer proteins (PITPs) transport PtdIns from the donor ER membrane to nuclear acceptor proteins such as p53. This transport process includes membrane docking, lipid extraction, membrane undocking, cytoplasmic diffusion, nuclear targeting, lipid deposition, and acceptor dissociation, followed by the PITP return to the donor membrane. Upon delivery of PtdIns to p53, the

(Figure legend continued at the bottom of the next page.)

phosphatase PTEN ensures precise regulation of nuclear AKT signaling by modulating PtdIns(3,4,5)P₃ turnover [16,102,109,110].

Recent studies demonstrate that the nuclear p53–PIP signalosome assembles the PtdIns(3,4,5)P₃ production complex by recruiting class I PITPα/β, PI4KIIα, PIPKIα, and IPMK, together with PDK1, mTORC2, and AKT [16,17,101,111]. This assembly activates the nuclear AKT pathway and promotes the phosphorylation of downstream targets, including FOXOs [16]. This nuclear p53–PIP–AKT signaling cascade enhances cancer cell survival, proliferation, migration, invasion, and chemoresistance. In addition, this pathway reveals a mechanism for *de novo* activation of nuclear AKT signaling within the non-membranous nucleoplasm, where protein scaffolds such as p53 assemble the PtdIns(3,4,5)P₃ lipid signaling complex [112].

Moreover, AKT can be activated in the nucleus by DNA-dependent protein kinase (DNA-PK) in a PtdIns(3,4,5)P₃-independent manner [113]. Under genotoxic stress, DNA-PK enhances AKT activity through two mechanisms: (i) by phosphorylating the SIN1 subunit of mTORC2 to promote AKT activation [114], and (ii) by directly phosphorylating nuclear-translocating AKT at Ser473, thereby inhibiting cisplatin-induced apoptosis [115].

As a tightly regulated kinase, AKT is inactivated by protein phosphatases, and Thr residues are dephosphorylated by protein phosphatase 2A (PP2A) in the cytosol and nucleus [116]. PP2A inhibitors, such as calyculin A and okadaic acid, maintain nuclear AKT activation induced by NGF treatment [117]. Immunoprecipitation shows an association between AKT and PP2A that peaks when nuclear AKT activity decreases [117]. The Ser residues of phospho-AKT are targeted by the PH domain leucine-rich repeat protein phosphatase (PHLPP) family of isozymes, including PHLPP1α, PHLPP1β, and PHLPP2, where PHLPP1β contains a bipartite NLS [118]. Although its role has not been established, nuclear PHLPP1 may be involved in regulating nuclear AKT.

Nuclear AKT targets a variety of substrates that control vital cellular processes. First, AKT regulates nuclear mTORC1 in response to growth factor signaling by phosphorylating RAN-binding protein 3 (RANBP3) and AKT1 substrate 1 (AKT1S1, or proline-rich AKT substrate 40 kDa, PRAS40) [119,120]. This phosphorylation facilitates RAPTOR nuclear translocation and relieves AKT1S1 inhibition of mTORC1, thus enhancing nuclear mTORC1 activity. Inhibition of nuclear AKT via the nucleus-specific AKT substrate-based tandem occupancy peptide sponge (AKT-STOPS) reduces nuclear mTORC1 activity, which regulates transcription by all three major RNA polymerases, and promotes cell survival and proliferation [119]. In addition to FOXOs, nuclear AKT phosphorylates the transcriptional coactivator p300 at Ser1834, enhancing its histone

complex undergoes sequential phosphorylation to generate PtdIns4P–p53, PtdIns(4,5)P₂–p53, and PtdIns(3,4,5)P₃–p53. This stepwise transformation requires the recruitment of PI-metabolizing enzymes, including phosphatidylinositol 4-kinase type IIα (PI4KIIα), phosphatidylinositol-4-phosphate 5-kinase type Iα (PIP5K), inositol polyphosphate multikinase (IPMK), phosphatase and tensin homolog (PTEN), and small heat-shock proteins (sHSPs). These events lead to the assembly of a PIP–p53 signalosome that recruits phosphoinositide-dependent protein kinase 1 (PDK1), mechanistic target of rapamycin complex 2 (mTORC2), and AKT, enabling nuclear AKT activation. In parallel, AKT can also be activated through a phosphoinositide-independent mechanism. Under genotoxic stress, DNA-dependent protein kinase (DNA-PK) enhances AKT activity by phosphorylating the stress-activated protein kinase-interacting protein 1 (SIN1) subunit of mTORC2 to promote AKT activation, and by directly phosphorylating serine 473 (Ser473) of nuclear-translocating AKT. Nuclear AKT is inactivated by protein phosphatases where Thr residues are dephosphorylated by protein phosphatase 2A (PP2A), which can be inhibited by calyculin A or okadaic acid. Ser residues of phosphorylated AKT are potentially targeted by nuclear PH domain leucine-rich repeat protein phosphatase 1 (PHLPP1). Once activated, nuclear AKT phosphorylates multiple substrates, including ALY/REF export factor (ALY), forkhead box protein O (FOXO), AKT1 substrate 1 (AKT1S1), RAN-binding protein 3 (RANBP3), histone acetyltransferase p300, the apoptosis-related protein acinus, and zyxin. These phosphorylation events support nuclear processes such as mRNA export, repression of proapoptotic gene transcription, RNA polymerase II activation, histone acetylation, and chromatin remodeling. In addition, the receptor tyrosine kinase AXL activates TANK-binding kinase 1 (TBK1) which phosphorylates AKT3 at threonine 305 (Thr305) and serine 472 (Ser472) in the cytosol, promoting the nuclear translocation of the TBK1–AKT3 complex. Within the nucleus, this complex stabilizes the epithelial–mesenchymal transition (EMT) transcription factor snail, thereby facilitating EMT and metastasis. These nuclear AKT signaling mechanisms collectively promote cell survival, proliferation, migration, invasion, EMT, metastasis, and chemoresistance. This figure was created using BioRender.

acetyltransferase activity and transcriptional function [121]. AKT also phosphorylates the nuclear factor acinus; this prevents its caspase-mediated cleavage and inhibits chromatin condensation, thus blocking apoptosis [122]. Another AKT target is zyxin; phosphorylation by AKT strengthens the interaction between zyxin and acinus, preventing chromatin condensation [123]. Last, AKT-mediated phosphorylation of ALY is required for its binding to PtdIns(3,4,5)P₃, and this facilitates mRNA export and links nuclear AKT to the regulation of gene expression [124].

Concluding remarks

This review highlights compartmentalized AKT signaling at the PM, endosomes, protein scaffolds, and within the nucleus. AKT also functions in the cytosol [10], mitochondria [125], mitochondria-associated ER membranes (MAMs) [126], ER-to-Golgi trafficking vesicles [127], and the exocyst complex [128] (Figure 1). For instance, TANK-binding kinase 1 (TBK1) phosphorylates AKT at Thr308/Ser473 at the exocyst and promotes oncogenic transformation [128]. The compartmentalized AKT signaling adds complexity to how cells respond to stimuli, enabling precise modulation of cellular responses.

AKT isoforms – AKT1, AKT2, and AKT3 – share core activation mechanisms but execute distinct functions depending on the compartment and context. AKT1 promotes FA disassembly via FAK phosphorylation and supports nuclear programs that enhance migration through FOXO degradation and p300 activation [16,34,35]. AKT2 activates RAC1 and CDC42, leading to remodeling of the cytoskeleton and promoting lamellipodia formation – a function amplified by its recruitment to WDFY2-positive endosomes [38–40,63,64]. AKT1 and AKT2 interact with protein scaffolds such as CLIPR-59 to coordinate cell survival and glycogenesis [83,86]. The role of AKT3 remains less well defined; some studies suggest that it has a minimal impact on cellular migration [129], whereas others report that AKT3 downregulation increases metastasis. Mechanistically, activation of the AXL receptor induces TBK1-mediated phosphorylation of AKT3 in the cytosol, enabling nuclear translocation and accumulation of the TBK1–AKT3 complex. This complex stabilizes snail, a key transcription factor for epithelial–mesenchymal transition (EMT) in the nucleus, and thus promotes EMT and metastasis [130].

Beyond canonical phosphorylation at Thr308/Ser473, AKT activity is finely tuned by diverse post-translational modifications (PTMs) in a compartment- and isoform-specific manner. Acetylation by p300 inhibits PtdIns(3,4,5)P₃ binding, whereas deacetylation by the NAD-dependent deacetylase sirtuin-1 (SIRT1) restores membrane recruitment [131]. On the membrane, SET domain bifurcated histone lysine methyltransferase 1 (SETDB1) trimethylates AKT at Lys64, enabling recruitment of E3 ubiquitin ligases such as tumor necrosis factor receptor-associated factor 6 (TRAF6) and S-phase kinase-associated protein 2 (SKP2), which catalyze Lys63-linked ubiquitination and activate AKT upon growth factor stimulation [132]. TRAF6 also mediates Lys8/Lys14 ubiquitination in the nucleus, facilitating return of AKT to the membrane and regulating G protein-coupled receptor signaling [133]. The E3 ligase tetratricopeptide repeat domain 3 (TTC3) also binds to phosphorylated AKT and targets it for nuclear degradation, potentially contributing to Down syndrome pathology [134]. Modification by addition of small ubiquitin-like modifier (SUMOylation) at Lys276 by protein inhibitor of activated STAT1 (PIAS1), that is reversed by SUMO-specific protease 1 (SEN1P), enhances nuclear AKT accumulation, cyclin D1 expression, and tumorigenesis [135,136]. O-linked N-acetylglucosamine (O-GlcNAc) modifications further modulate AKT: O-GlcNAcylation at Thr305/Thr312 disrupts the AKT–PDK1 interaction, impairs phospho-Thr308, and suppresses proliferation [137], whereas modification at Thr430/Thr479 enhances phosphorylation and contributes to diabetic vascular calcification [138].

Additional phosphorylation events refine AKT structure and activity. mTORC2-mediated phosphorylation at Thr450 supports proper AKT folding [139]. Phosphorylation at Ser477/Thr479

Outstanding questions

How do phosphatases and PTMs coordinate AKT activity? How do phosphatases act in a compartment- and isoform-specific manner, and how do modifications such as ubiquitination, acetylation, and SUMOylation interact with canonical phosphorylation to modulate AKT signaling?

How is AKT compartment-specific activation regulated? What molecular mechanisms govern its activation at membrane microdomains such as lipid rafts and endosomes, and how do these spatial cues shape downstream cellular responses?

How do protein scaffolds shape the specificity of AKT signaling? How do scaffolds integrate AKT with other signaling pathways to fine-tune outcomes such as cellular migration, proliferation, and survival in both normal and disease contexts?

What regulates AKT nuclear translocation and function? What mechanisms control AKT import into the nucleus, and how does nuclear AKT regulate chromatin remodeling, gene expression, and chemoresistance?

How can innovative tools improve our understanding of spatial AKT signaling? What advances in biosensors, live-cell imaging, or *in vivo* models will be necessary to visualize AKT activity and phosphoinositide dynamics with spatial and temporal precision?

Can spatial targeting of AKT signaling improve cancer therapy? Could selective modulation of AKT activity in specific subcellular compartments enhance therapeutic efficacy while minimizing systemic toxicity in precision oncology?

by cyclin-dependent kinase 2 (CDK2), mTORC2, or DNA-PK primes AKT for full activation in response to growth, DNA damage, or mitogenic signals, depending on its subcellular localization [140]. Many of these PTMs exhibit isoform preferences; for example, SETDB1 more strongly methylates AKT1 than AKT2/3 [132].

Spatially coordinated AKT signaling governs key cellular processes and contributes to oncogenesis. During migration, PM-activated AKT drives membrane protrusions and endocytosis, endosomal AKT sustains signaling to the nucleus, and nuclear AKT regulates EMT and cytoskeletal remodeling. Under stress, nuclear and cytoplasmic AKT collaborate to suppress apoptosis. Disruption of this spatial signaling underlies disease progression and therapy resistance. Compartment-specific AKT isoforms further define oncogenic outcomes: nuclear AKT1 promotes chemoresistance via p53–PIP signalosomes, endosomal AKT2 drives metabolic reprogramming in obesity-linked cancers, and nuclear AKT3 facilitates metastasis through EMT.

Despite progress, key challenges remain (see [Outstanding questions](#)). The compartment-specific functions of AKT isoforms, especially AKT3, are not fully understood. The roles of PTMs, phosphatases, and scaffolds in shaping spatial signaling require deeper mechanistic insight. Technical limitations in visualizing real-time PIP dynamics and AKT activity *in vivo* also hinder advancement. Addressing these gaps demands new tools – such as advanced biosensors and organelle-targeted inhibitors – and refined model systems.

In summary, this review reframes AKT signaling beyond the canonical membrane-centric model by integrating compartmental networks, isoform divergence, and PTM complexity. Targeting compartment- and isoform-specific AKT functions offers a promising path to overcome resistance and advance precision oncology. Realizing the full potential of AKT-targeted therapies requires a shift from broad inhibition to spatially and context-specific strategies – such as inhibition of nuclear effectors (e.g., DNA-PK, IPMK, or TBK1) or disrupting endosomal scaffolds – to improve efficacy and reduce resistance.

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Declaration of interests

The authors declare no competing interests.

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